LAB TASK-03,

ROLL NO: 25K-2025

1. Write a program to create an Electricity Bill Calculator which will ask the user for Customer Name, Units Consumed (value should be of integer), Price per Unit (value should be of float) and calculate the Total Bill and display it in a formatted bill slip.

SOURCE CODE:

```
#include<stdio.h>

int main(void){
    char c_n[50];
 int u_c;
 float Price_per_Unit,bill;
 /* c_n= customers name, u_c=units consumed*/

 printf("Enter your name:");
 scanf("%s",&c_n);

 printf("\nEnter the total units consumed:");
 scanf("%d",&u_c);

    printf("\n Whats the price per unit?:");
    scanf("%f",&Price_per_Unit);

  bill=u_c*Price_per_Unit;

  printf("\n--------------------------------");
   printf("\n\t ELECTRICITY BILL");
    printf("\n------------------------------");
    printf("\nCustomer:%s", c_n);
printf("\nUnits Consumed:%d", u_c);
printf("\nPrice per Unit:%.5f", Price_per_Unit);
printf("\n-----------------------------");
printf("\nTotal Bill: %.2f PKR\n", bill);
printf("-----------------------------");

return 0;
}
```

RESULT:

```
Enter your name:Aliza

Enter the total units consumed:270

 Whats the price per unit?:12.5

---------------------------------
         ELECTRICITY BILL
-------------------------------
Customer:Aliza
Units Consumed:270
Price per Unit:12.50000
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Total Bill: 3375.00 PKR
-----------------------------
--------------------------------
Process exited after 8.31 seconds with return value 0
Press any key to continue . . .
```

2. Write a program to create a Shopping Receipt Generator which takes as input eggs, bread and butter prices as float with 2 decimal places and then generate a receipt showing each item’s price, subtotal, 17% sales tax and final total.

SOURCE CODE:

```c
#include <stdio.h>

int main(void) {
    float eggs, bread, butter;
    float subtotal, tax, final_total;

    printf("Enter price of eggs: ");
    scanf("%f", &eggs);

    printf("Enter price of bread: ");
    scanf("%f", &bread);

    printf("Enter price of butter: ");
    scanf("%f", &butter);

    subtotal = eggs + bread + butter;
    tax = subtotal * 0.10;
   final_total = subtotal + tax;

    printf("\n-------------------------------\n");
    printf("        SHOP RECEIPT\n");
    printf("-------------------------------\n");
    printf("Item 1: %.2f\n", eggs);
    printf("Item 2: %.2f\n", bread);
    printf("Item 3: %.2f\n", butter);
    printf("-------------------------------\n");
    printf("Subtotal: %.2f\n", subtotal);
    printf("Tax (10%%): %.2f\n", tax);
    printf("-------------------------------\n");
    printf("Grand Total: %.2f\n", final_total);
    printf("-------------------------------\n");

    return 0;
            }
```

RESULT:

```
Enter price of eggs: 40
Enter price of bread: 15
Enter price of butter: 20

-------------------------------
        SHOP RECEIPT
-------------------------------
Item 1: 40.00
Item 2: 15.00
Item 3: 20.00
-------------------------------
Subtotal: 75.00
Tax (10%): 7.50
-------------------------------
Grand Total: 82.50
-------------------------------


--------------------------------
Process exited after 9.871 seconds with return value 0
Press any key to continue . . .
```

3. Write a program to create a Fuel Consumption Tracker that calculates car's efficiency and takes input distance travelled (km, float), fuel used (litres, float), fuel price per litre, fuel price per litre(float) and gives output distance travelled, fuel consumption (km per litre with 2 decimal places) and total fuel cost.

SOURCE CODE:

```c
#include <stdio.h>

int main(void) {
    float distance, fuel, pricePerLitre;
    float efficiency, totalCost;

    printf("Enter distance (km): ");
    scanf("%f", &distance);

    printf("Enter fuel used (liters): ");
    scanf("%f", &fuel);

    printf("Enter fuel price per liter: ");
    scanf("%f", &pricePerLitre);

    efficiency = distance / fuel;
    totalCost = fuel * pricePerLitre;
    printf("-------------------------------\n");
    printf("FUEL REPORT\n");
    printf("-------------------------------\n");

    printf("Distance: %.1f km\n", distance);
    printf("Fuel Used: %.1f L\n", fuel);
    printf("Efficiency: %.2f km/L\n", efficiency);
    printf("Total Cost: %.2f PKR\n", totalCost);
    printf("-------------------------------\n");

    return 0;
}
```

RESULT:

```
Enter distance (km): 50
Enter fuel used (liters): 36.7
Enter fuel price per liter: 100
-------------------------------
FUEL REPORT
-------------------------------
Distance: 50.0 km
Fuel Used: 36.7 L
Efficiency: 1.36 km/L
Total Cost: 3670.00 PKR
-------------------------------

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Process exited after 41.52 seconds with return value 0
Press any key to continue . . .
```

4. Create a Student CGPA Calculator that takes marks of your 1stsemester's subjects (float), each subject is out of 100. Calculate percentage and convert percentage into CGPA (out of 4.0 scale) using the formula CGPA = (Percentage/100) * 4 and display with proper formatting.

<u>SOURCE CODE:</u>

```c
#include <stdio.h>

int main(void) {
    float sub1, sub2, sub3, sub4, sub5;
    float total, percentage, cgpa;

    printf("Enter marks of 5 subjects: ");
    scanf("%f %f %f %f %f", &sub1, &sub2, &sub3, &sub4, &sub5);

    total = sub1 + sub2 + sub3 + sub4 + sub5;
    percentage = (total / 500.0) * 100;
    cgpa = (percentage / 100.0) * 4.0;

    printf("-------------------------------\n");
    printf("STUDENT RESULT\n");
    printf("-------------------------------\n");

    printf("Marks: %.1f, %.1f, %.1f, %.1f, %.1f\n",
           sub1, sub2, sub3, sub4, sub5);
```

```
    printf("Total: %.1f\n", total);
    printf("Percentage: %.2f%%\n", percentage);
    printf("CGPA: %.2f / 4.00\n", cgpa);
    printf("-------------------------------\n");

    return 0;
}
```

RESULT:

```
Enter marks of 5 subjects: 70 60 50 40 80
-------------------------------
STUDENT RESULT
-------------------------------
Marks: 70.0, 60.0, 50.0, 40.0, 80.0
Total: 300.0
Percentage: 60.00%
CGPA: 2.40 / 4.00
-------------------------------

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Process exited after 11.1 seconds with return value 0
Press any key to continue . . .
```

5. Write a program to create a Loan EMI Calculator that calculates the monthly EMI for a loan and takes input the Loan Amount (float), Annual Interest Rate (float in %), Loan Duration (years, integer) and formula for calculating EMI is EMI = [P × r × (1 + r)^n] / [(1 + r)^n – 1], where P = Principal Loan Amount, r = Monthly Interest Rate = Annual Rate/(12*100), n = Total Months = Years * 12.

SOURCE CODE:

```c
#include <stdio.h>
#include <math.h>

int main(void) {
    float P, annualRate, monthlyRate, EMI;
    int years, n;

    printf("Enter Loan Amount: ");
    scanf("%f", &P);

    printf("Enter Annual Interest Rate (%%): ");
    scanf("%f", &annualRate);

    printf("Enter Duration (years): ");
    scanf("%d", &years);

    monthlyRate = annualRate / (12 * 100);
    n = years * 12;
    EMI = (P * monthlyRate * pow(1 + monthlyRate, n)) /
          (pow(1 + monthlyRate, n) - 1);

    printf("-------------------------------\n");
    printf("LOAN CALCULATION\n");
    printf("-------------------------------\n");
    printf("Loan Amount: %.2f\n", P);
    printf("Duration: %d years (%d months)\n", years, n);
    printf("Interest Rate: %.2f%% per year\n", annualRate);
    printf("Monthly EMI: %.2f\n", EMI);
    printf("-------------------------------\n");

    return 0;
}
```

RESULT:

```
D:\UNIVERSITY!!!\c programs'

Enter Loan Amount: 1000000
Enter Annual Interest Rate (%): 10
Enter Duration (years): 5
-------------------------------
LOAN CALCULATION
-------------------------------
Loan Amount: 1000000.00
Duration: 5 years (60 months)
Interest Rate: 10.00% per year
Monthly EMI: 21247.06
-------------------------------


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Process exited after 11.62 seconds with return value 0
Press any key to continue . . .
```